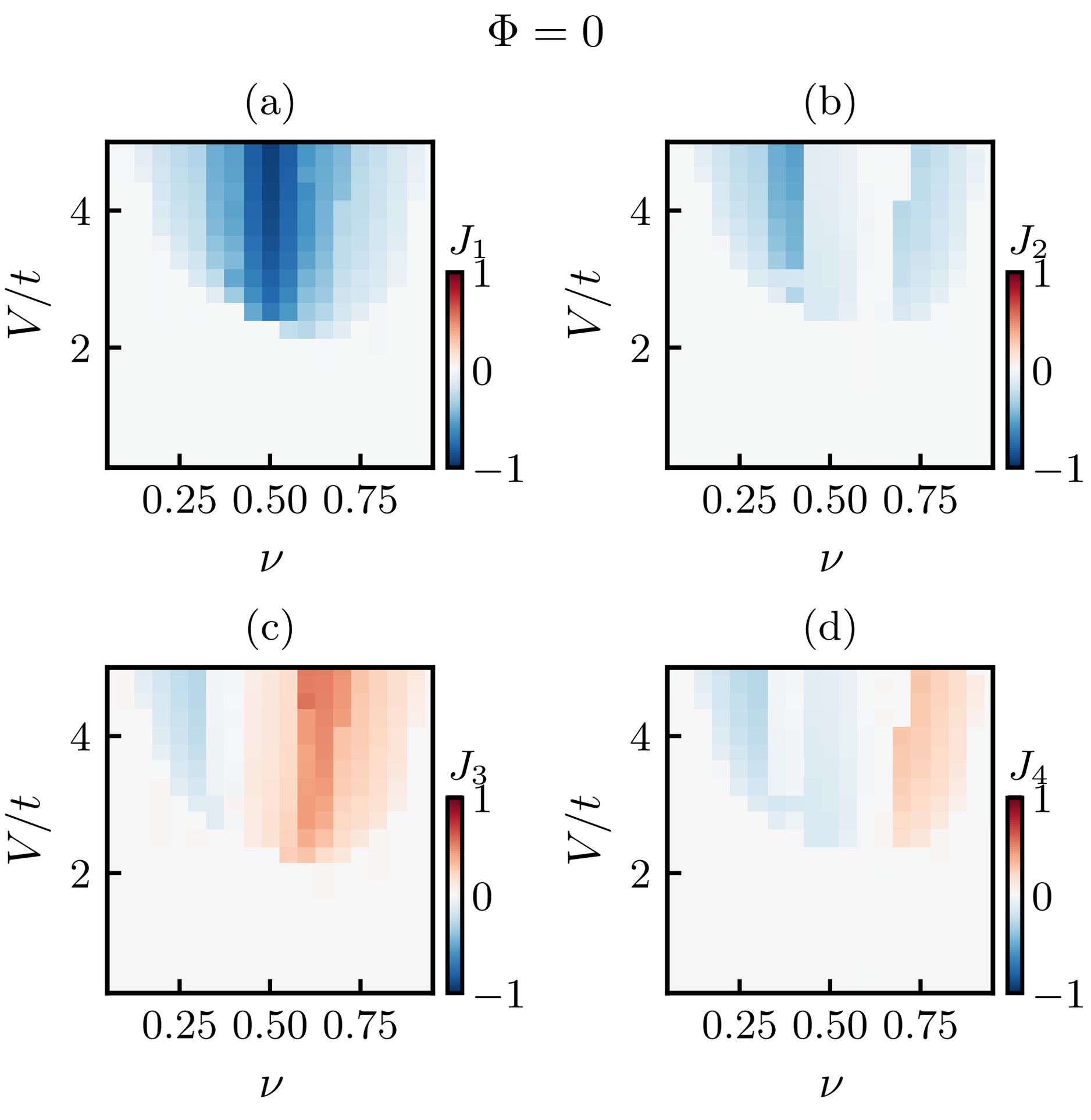

$\Phi = 0$
(a)
(b)
(c)
(d)
V/t
ν
4
2
0.25 0.50 0.75
J_1
J_2
J_3
J_4
1
0
−1